Exploring Evolutionary and Structural Dynamics of β-Galactosidases: A Comparative Genomics Approach Highlighting Thermophilic Motifs

Abstract: This study investigates the evolutionary and structural aspects of β-galactosidases through a comprehensive analysis of sequence motifs, phylogenetic distribution, and structural models. It was identified β-galactosidase sequences and characterized conserved motifs using Pfam and CAZy domain annotations from KEGG and CAZy databases. Also, structural comparisons were performed using PDB models, and phylogenetic analyses were conducted to assess evolutionary relationships among enzyme variants. Additionally, we explored amino acid content across sequences, particularly focusing on motifs associated with hyperthermophilic organisms, through custom Python and R scripts. By using a comparative hybrid method, it was possible to recognize at least three different groups, according with the primary structure of motifs and shared structure for the groups, describing a wide diversity from those enzymes involving into similar reactions. Our findings shed light on the evolutionary history, functional diversity, and potential biotechnological applications of β-galactosidases.

Keywords: β-galactosidase, comparative genomics, conserved motifs, structural analysis, phylogenetics, amino acid content, hyperthermophiles, biotechnological applications.

Introduction

The study of complete genomes and meta-genomics data has broadened our understanding of metabolic pathways, revealing conserved elements, alternate enzymes, and novel genes and metabolic routes (Collins et al., 2011). Comparative genomic analysis and exploration of environmental samples have led to the discovery of new branches in the Archaea phylogenetic tree (Zaremba-Niedzwiedzka et al., 2017), insights into membrane conformation and metabolism,

including glycerol uptake (Schönknecht et al., 2013), and the recognition of vesicular transport as an ancient feature originating from early Archaean lineages (Liu et al., 2021). By comparing genomes and analyzing meta-genome studies, we can identify features shaping lifestyles, adaptations, and gene evolution, revealing conserved elements and their roles in genomic arrangements even in both Bacteria and Archaea (Nei & Rooney, 2005). To understand these relationships provides insights into the emergence of new metabolic pathways and their connections to common ancestors.

The concept of a gene family is pivotal in studying, comparing, and organizing genes into a phylogenetic framework of duplication and diversification. Originating from Ohno's idea, a gene family comprises genes inferred to have evolved from duplication and divergence events (Henikoff et al., 1997; Innan & Kondrashov, 2010). Each gene family associates a single gene with a representative organism, implying shared ancestry among genes and similar functions across closely related species, termed orthologs (Henikoff et al., 1997; Innan & Kondrashov, 2010).

However, some genes exhibit uncertain orthology due to close homology and diverse functions, categorized as nonorthologous elements. These genes, termed paralogs, result from specific gene duplication events in certain representatives (Demuth & Hahn, 2009; Eirín-López et al., 2012). Gene families are often defined by the arrangement and presence of conserved motifs, linking them with specific folding features and functional properties, including interactions with cofactors and structural arrangements among subunits (Magner et al., 2015).

For instance, exploring one of the most studied hydrolases, beta-galactosidase, begins with examining the gene encoding its most representative form documented in the literature, the lacZ beta-galactosidase.

This enzyme has three enzymatic activities: *i)* it can degrade the disaccharide lactose to form glucose and galactose. *ii)* the enzyme can catalyze the transgalactosylation of lactose to allolactose, and *iii)* the allolactose can be cleaved to the monosaccharides.

It is allolactose that binds to *lacZ* repressor and the defined lac operon that regulates their transcription due to the repressor enzyme interaction (Juers et al., 2012). It is well known the structure of this enzyme. lacZ beta-galactosidase is a tetramer of four identical polypeptide chains. And this structure is defined by is fold in five different domains. The central domain resemble to the triose phosphate isomerase (TIM) folding that comprises a structure of α8β8 domain forming an active site. However is possible to recognize enzymes involved in similar reactions , but their evolutionary or structural homology is still not studied, by a focus of gene families or included into a structural comparison.

There is three distinct lineages: *ebgA*, *bgaB*, and *lacA*, of galactosidades, with the same ~~EC number~~ (Kanehisa, 2000). The early comparison of beta-galactosidases suggests their divergence either in the context of Eukarya or through specific duplication and divergence events within certain phylogenetic groups (Kondrashov et al., 2002; Konrad et al., 2011).

The complexity of gene families, particularly beta-galactosidases exhibit a diverse range of probable homologs despite sharing a catalytic motif (Wang et al., 2014). Gene families often undergo subfunctionalization, neofunctionalization, and subsequent gene fusion events following early duplication events. This phenomena was a process widely distributed in the evolutionary history of comparative genomics in eukaryotic groups, and in recent studies is still described in proakryotic taxonomic levels and representatives (Gogarten & Olendzenski, 1999).

Moreover, the presence of certain motifs can influence structural elements, affecting enzyme specificity, subunit interactions, and overall functional states. Research efforts focus on integrating

motif recognition across various databases to understand these structural-functional relationships more comprehensively. ((Becerra et al., 2007; Thoma et al., 1998)Thoma, 1998, Becerra, et al. 2007). One strategy to recognize these kind of shared elements in order to recognize their properties to infer probable convertgences and homologous elements is to study the enzymes at tertiary structure level (Russell & Barton, 1992(Russell & Barton, 1992)).

Tertiary structure comparison based on bioinformatics models and theoretical tools to infer conserved regions and fucntional implications linking particular structures with motifs and the importance of conserved domains. The inference of these comparisons combine the recognition of particular element that bring particular role ~~since~~ the activity and cofactor relation in the catalytic side to even those amino acids whose hydrophobic/hydrophilic interaction bring consequences among the different subunits and interaction with other proteins(Dunker & Lathrop, 1995)

One of the most significative ~~effords~~ to recognize and order the study of the proteins is to link them into protein families and ~~strcutral~~ homologues where,according with the conserved motifs, structural coincidence and tri-dimensional alignment and comparing its function and since the order to the ~~linbk~~ and possible phylogenetic relation between structural elements bring particular elements in comparison studies, first, to recognize its enzymatic reaction and second to recognize particular arrangements among the different chains to have unique or shared functions among the subunits to bring its holozyme active structure (Pandit et al., 2002; Shafee et al., 2016)

According with the actual way to compare and define an structural homologues among the wide diversity that the databases present, two strategies are taken according with the particular elements to take in the analysis, those strategies are defined by the RMSD and the TM-score

The ~~root-mean-square deviation~~ (RMSD) serves as an intuitive metric, fulfilling the criteria of a metric. Alternatively, interatomic distance matrices can be compared, yielding a difference measure in distance space. These measures are specialized and applicable not only for pairwise comparisons

but also for comparing groups of structures. A useful measure for comparing the 'spread' within a complete group is to compute the average RMSD of all pairs of structures, employing Kabsch's formula for pairwise comparisons. The resulting covariance matrix, based on an isotropically distributed ensemble (IDE), integrates information to identify which structure is closest to the average value of the compared group (Brüschweiler, 2002). However, while RMSD provides a metric for structural similarity, it may be influenced by a small set of divergent atoms in loop regions, as globally optimal transformations minimize distances between all superimposed atom pairs (Li, 2013).

Recent studies suggest that RMSD is not solely dependent on the overall fit quality but also on protein length (Hung & Samudrala, 2012). Considering beta-galactosidase's relatively short amino acid sequence, alternative measures for comparison are warranted. Cuneo et al.'s work comparing a thermophilic enzyme's tertiary structure to a mesophilic one highlights a shared folding pattern, based on the docking and enveloping of approximately 85% of the alpha carbons of the backbone (Cuneo et al., 2008).

The ~~TM-score~~, a counterpart measure to RMSD, employs a variant of the Levitt–Gerstein (LG) metric, providing a length-independent measure that mitigates the impact of divergent pairs of atoms in superimposed structures (Xu & Zhang, 2010). Unlike RMSD, TM-score does not have a straightforward relationship between the covariance matrix and the optimal transformation that maximizes LG. Instead, it applies the Kabsch algorithm to superimpose different subsets of atoms and measures the LG score over the entire protein. Despite its effectiveness, the TM-score algorithm is slower due to the need for multiple tries and superposition events (Hung & Samudrala, 2012).

TM-score facilitates comparison and evaluation of additional branches in enzyme structure configuration. Xu and Zhang suggest a TM-score greater than 0.5 indicates the same fold in pairwise or multiple alignments, guiding recognition of structural homologs within similar groups.

Comparing structural elements aids in inferring conserved regions and properties, especially in beta-galactosidase groups where structural homologs may vary, such as homotetramers (Uchil et al., 2017). Incorporating previously reported motifs enhances structural comparison accuracy.

In recent studies, orthologous gene groups are often defined through comparative genomics across multiple organisms (Koonin, 2005). Comparing genome sequences and assessing similarity enables the identification of shared genes within a group (Demuth & Hahn, 2009). These genes may trace back to a hypothetical common ancestor of the studied samples (Kristensen et al., 2011).

The studies by comparing particular features of the enzymes,,their prevalence in certain motifs and even its particular structure it have been a strategy in order to recongize particular elements whose can be conserved in to certain protein families, being this proposal to recognize a groupof sequences and/or structures involved into a shared evolutionary story among all the sequence (Finn et al., 2010; Holm, 2022).The extension of this proposal can be extrapolated since its particular reaction to its structural distribution and arrangement. The prevalence of this proposar was recognized into beta- galactosidases, a particular group defined in previous references that enable to even define shared and divergent elements (Anisha, 2016)

The enzymes associated with beta-galactosidases, despite their diversity, may share a common evolutionary path. Even within specific phylogenetic bacterial groups, the diversity observed presents an opportunity to identify recent diversification or conserved elements that could shed light on the diversification of the lacZ gene or the occurrence of convergence and substitution in different phylogenetic representatives.

One crucial aspect to consider is the study and prior relationships established in specific databases vital for comparing different enzymes and their corresponding genes. Two such databases relevant to this study are the KEGG database and the CAZy database.

The KEGG database, originally conceived as an open-access resource for biomedical and biochemical research, now plays a crucial role in identifying homologous genes, particularly those recently sequenced, through blast-based searches and recognition of coding sequence elements. To enhance classification and analysis within this database, methods such as integrating Pfam have been employed (Mistry et al., 2021).

Pfam is a widely used classification method that breaks down protein structures into specific sequence domains, aiding in the identification of shared domains in both KO groups and gene family databases. Developed over many years with extensive manual curation by domain experts, this approach is highly accurate and practical. By analyzing primary structure sequences, Pfam can identify alignments and potential functions of sequence regions (Finn et al., 2010a; Mistry et al., 2021).

In this study, Pfam is utilized to enhance comparison and classification efforts, aiming to establish robust and comprehensive groups of homologous elements among beta-galactosidases. ~~The objective~~ is to identify potential evolutionary paths and arrangements for different beta-galactosidase KO groups by combining conserved motifs, matched tertiary structures, and recognizing conserved regions in protein structures. Additionally, the study seeks to infer phylogenetic relationships among orthologous groups beyond the EC number, complementing the recognition and study of distant orthologs, at least defined by TIM barrel regions (Chan et al., 2017).

Utilizing their metabolic background, complete metabolic pathways can be defined within complete genomes, identifying punctual absence and presence in the proteomic reports of specific organisms with previously reported complete genomes. This approach allows for the identification of interaction networks, links among different databases, and a set of KO orthologous groups, providing an initial understanding of completely reported genes involved in orthologous groups

based on sequence homology and the shared presence of Pfam motifs to support conserved regions with functional properties (Kanehisa, 1996, 2000; Kanehisa et al., 2004).

This study is focused into explore and establish the equivalence between the KEGG and CAZy databases regarding KO groups associated with beta-galactosidase. By conducting a comparative analysis, the study aims to identify differences and similarities in the sequences reported in both databases.

Furthermore, the study seeks to identify particular features that could enable the proposal of an alternate scheme, combining tertiary structure models with conserved or derived elements. This involves comparing KO Pfam and structurally related elements with data obtained from the CAZy database.

The CAZy database focuses on reclassifying genes involved in carbohydrate-related metabolism, aiming to enhance understanding of the distribution and relationships among different protein families involved in assembling and breaking down complex carbohydrates (Drula et al., 2022). Unlike traditional approaches, the construction of this database involves multiple analyses, comparisons, and the implementation of bioinformatic algorithms, resulting in a comprehensive understanding of relationships among newly reported sequences. Although CAZy is a recent proposal, its impact on defining homology roles in newly sequenced databases is significant, particularly in conjunction with the Kyoto Encyclopedia of Genes and Genomes (KEGG).

To define conserved features and important properties of beta-galactosidases, we'll combine data from KEGG and CAZy databases with structural analyses. This involves:

a) **Database Integration**: We'll use KEGG and CAZy data to identify beta-galactosidase KO groups and associated sequences, providing insights into their evolutionary history and functions.

b) **Structural Comparison**: We'll compare crystal structures and employ new methods like computational modeling to understand different arrangements and functional elements of beta-galactosidases.

c) **Sequence Analysis**: By aligning sequences and analyzing motifs, we'll identify conserved features, distinguishing between paralogous and homologous components.

d) **Functional Characterization**: We'll investigate functional properties such as substrate specificity and catalytic activity, providing context for understanding beta-galactosidase diversity.

This approach will offer a comprehensive view of beta-galactosidase evolution, function, and structure, revealing shared evolutionary histories and functional diversity across organisms.

Furthermore, leveraging structural data allows us to compare and expand relationships among various enzymes with galactosidase activity and assess their distribution.

Lastly, within the KO/CAZy groups, we'll investigate if representatives with extremophilic changes in sequence or structure are present. We'll evaluate the significance of these changes compared to the conserved group signal, employing cluster analysis and comparing structural/Pfam motifs to identify particular elements involved.

Methods

To achieve the study and comparison in beta-galactosidases, a comparative focus was applied using the KEGG and CAZy databases based in keywords, obtained since sequence information, detecting structural c~~o~~served motifs and detecting possible shared features. The KEGG database (Kanehisa, 2000) (https://www.genome.jp/kegg/) was used to obtain sequences related to the enzymatic capacity of beta-galactosidase based in keywords and particular enzyme reactions based in the EC

number Each collected sequence was annotated with its average protein sequence, KEGG ID code, EC number, and definition. Additionally, to enhance sequence characterization, the Pfam, a tool that allows for the identification of conserved protein domains was used.

To further characterize the sequences and explore possible elements for comparison, a Python script was developed to extract species names from each sequence header and analyze the distribution of sequences across different kingdoms. These scripts are available on our GitHub repository (https://github.com/hectorianus1977/pythonforbioinformatics), providing support for other bioinformatics analyses.

Additionally, the distribution of Pfam values obtained from the KEGG database was confirmed in the Pfam/InterPro database (Finn et al., 2010) (http://pfam-legacy.xfam.org/). Significantly, motifs with an e-value lower than 10e-15 were considered, characterizing specific regions in the sequences and structural models. Furthermore, to select representative examples for comparison, the Protein Data Bank (PDB) was consulted to identify homologous sequences and structural models.

In order to identify Carbohydrate-Active Enzyme (CAZy) domains among the KO group sequences, a stand-alone HMMER search was performed (Finn et al., 2015) across all the retrieved β-galactosidase sequences. The reference profiles were downloaded from the dbCAN-HMM database (Version 11, release date: 2022-08-09) hosted at the dbCAN3 server (URL: https://bcb.unl.edu/dbCAN2/index.php) (H. Zhang et al., 2018). For proper CAZyme annotation, the raw output was parsed and filtered with a coverage threshold of 0.5 and an E–value threshold of 1×10^-5.

The results of this search were captured in a CSV file to recognize the diversity of motifs and conserved regions. This resulting file is added into the github place in order to share the resulting database. Additionally, to compare a particular tertiary structure model representing and probing the conservation of structure and recognizing Pfam conserved motifs, a search with at least 5

different random sequences was conducted in the Protein Data Bank (PDB) to select the most characterized PDB model. Using the EC number for comparison, structures were compared using the background of Pfam and the structure comparison tool on the PDB webpage, specifically the pairwise alignment viewer (Zardecki et al., 2022), utilizing the FATCAT algorithm in flexible mode (Ye & Godzik, 2004). This allowed for the evaluation of identity and similarity regions, TM values, and the recognition of homologous and conserved elements.

The grouped data were presented in the corresponding figure, where, according to the analysis, probable homology among structures could be inferred. For example, due to the incidence of a sole CAZy motif, it was possible to define that K01190 and K12111 KO groups might be homologous because of similar motifs, GH2. Additionally, the comparison was further supported with structural models, allowing for a comparison with the PDB pairwise comparison tool, to represent both Pfam motifs and their structural counterparts only to confirm the probable homology among these structures who representa KO group.

To infer evolutionary relationships and identify conserved motifs within β-galactosidase sequences, we constructed a Maximum Likelihood phylogenetic tree using the PRALINE multiple sequence alignment algorithm (Simossis & Heringa, 2005). This approach aligns sequences with preferences in motif recognition, allowing for the identification of shared signals across sequences. The IQ-TREE shell version was utilized in combination with PRALINE to propose a phylogenetic distribution, aimed at recognizing shared arrangements and elements across all groups.

For all comparisons, representative arrangements recognized in the CAZy database using the HMMER algorithm (Wong et al., 2011) were used. This classification was prioritized due to its background in recognizing particular motifs associated with thermo-tolerance or specific phylogenetic groups represented in sequences obtained from thermophilic/hyperthermophilic microorganisms and different KO groups.

The resulting analysis enabled the linking of enzymes belonging to organisms with thermophilic/hyperthermophilic lifestyles, or at least recognition of any particular features reported in Pfam or CAZy that resemble a diverse or common signal of motifs. Multiple elements were detected, providing insights into the diversity and evolutionary relationships of β-galactosidases.

Furthermore, after characterizing the different models defined by motif structure, we sought to group sequences and differentiate them based on motifs, or infer possible amino acids that bring optimal variance for mesophilic/thermophilic species. This was achieved by comparing amino acids that exhibit variance among closely related phylogenetic species with differing thermophilic properties. These analyses will be presented within the relevant KO groups where applicable, providing insights into the functional and evolutionary aspects of β-galactosidases.

The methodology employed to compare and identify potential homologies among the KO groups involves maintaining specific manual filters. These filters compare elements from Pfam, CAZy, and even sequence alignments, focusing on the distribution and homology of particular motifs and functional elements. This approach has enabled us to discern that despite the genetic diversity present within group K12308, as indicated by its complex enumeration in Table 1, a common component can still be identified, particularly around the N-terminal and central regions of the sequences.~~Pasar a materiales y metodos~~

To visualize the Pfam or CAZy motifs in the tertiary structure model, the free software Pymol on the Linux platform was used. This allowed us to assign differential colors based on motif length and distribution. Specifically, we focused on Chain A for each case to highlight and compare the motifs.

For cases where the motif was implicated in a unique distribution related to sequences obtained from extremophilic/hyperthermophilic organisms, we employed additional software tools. Upon identifying regions potentially involved in adaptations to extreme lifestyles, sequences were

selected and arranged into a fasta file. Using PRALINE, common regions and derived properties of possible duplicated sequences were recognized.

To infer the distribution of each conserved region in the tertiary structure model, we utilized UCSF Chimera software. The match function within UCSF Chimera allowed for the integration of conserved regions into the tertiary structure model using the alignment file and the pdb file.

To further explore an alternative approach to relating amino acid content across all analyzed sequences, we combined the results of custom Python scripts and R scripts to evaluate and compare amino acid content, particularly focusing on motifs originating from sequences obtained from hyperthermophilic organisms.

The Python script facilitated the counting and transformation of obtained values into a common scale to compare amino acid content across all sequences. Subsequently, an R script was utilized to conduct cluster analysis using hierarchical clustering with the ward method, generating a dendrogram that evaluated amino acid content rather than the specific positions within these sequences.

The resulting dendrogram provided insights into the similarity of amino acid content across sequences, offering potential avenues for substitution and refinement in future biotechnological studies and modeling efforts. The R script utilized included code corresponding to the ~~hclust~~, factoextra, and cluster packages, sourced from the reference guide related to these elements (Kassambara, 2017). All partial files were read using notebook mode in the Linux Ubuntu desktop.

Results

The primary outcome of the initial search was the identification of nine groups of orthologous genes in KEGG database. The search and compilation of all sequences were not exhaustive due to the limitations of the KEGG database, which tends to overestimate the number of gene copies and

sizes. The database categorizes diverse genes and proteins solely based on their membership in orthologous groups, even if some gene copies only share major Pfam motifs and lack other C-terminal motifs crucial to the gene's overall structure.

To discern individual unique copies, we analyzed the structure and reported Pfam motifs of each KO group. These details are summarized in Table 1. From this preliminary analysis, we identified that eight of the nine groups are associated with beta-galactosidase function. The exception, group K12112, is classified as a class IV enzyme, this group was excluded from further study because its elements had been extensively analyzed by Hall (1999), and its partial classification suggests it is a recent addition exclusive to Gammaproteobacteria. The remaining eight groups underwent detailed analysis concerning prevalence and connections between two or more groups.

Moreover, to understand the distribution of these groups focused on a prokaryotic and possibly extremophilic lifestyle might be present, we categorized each sequence according to cellular domain involvement, as shown in Figure 1.

The key finding involves the basic classification performed automatically by a Python script, which utilized a dictionary of species to count the collected sequences manually. This process allowed us to discern that groups K12309, K18579, and K25543 exclusively contain sequences from eukaryotic genomes. The other five groups predominantly include sequences from various prokaryotic taxonomic groups, with homologs also identified in the Archaeal domain for groups K01220 and K01224. K01190 also include Plantae and Fungi.

To highlight sequences solely associated with prokaryotic contexts, we noted universal distribution across groups K01190, K01220, K01224, K12111, and K12308. Notably, almost all groups featured representation from Gammaproteobacteria, except K01220. This overrepresentation of Gammaproteobacteria, encompassing strains and varieties of *Escherichia* spp. and *Salmonella* spp., was observed across the KO groups. However, their distribution will not be prioritized in

comparative studies. Conversely, the Bacillota group is predominantly represented in K01220 and K01224. Although not shown in the accompanying graphic, this was a focused selection handmade for groups where Bacillota was particularly prevalent. ~~Esta selección manual esta en materiales y métodos?~~

Integrating the data presented in Table 1, KO groups K01190, K12111, K12308, and K12309 catalyze the same reaction and yield similar products (EC: 3.2.1.23). However, due to differences in structural arrangements and composition, we were able to distinguish and separately analyze these KO groups. Specifically, we divided them into two sets: one comprising K01190 and K12111, and the other consisting of K12308 and K12309.This division, rather than a unified comparison across all four groups, was justified by differences in their Pfam distributions and a preliminary structural comparisons. The distribution of conserved domains presented in each image repre~~n~~sent particular shared features that have a meaning into functional an structure the particular distribution of Pfam domains and shared structrure enable to separe in at least two different groups to compare .

In Figure 2, we inferred a shared functional signal between K01190 and K12111, highlighted by three Pfam motifs associated with glycohydrolases from the N- and C-terminals. Additionally, the CAZy database supports the conservation of the α8β8 barrel domain in these proteins, which features distinct N- and C-terminal motifs. A unique motif, only characterized in the K01190 representative structure (PDB: 3J7H), links to the lacZ enzyme structure through homology and possibly represents a duplication of the Bgall_small_N domain. Although these two motifs maintain similar structures in the K12111 representative (PDB: 6S6Z), the correlation is not completely straightforward. This is due to differences in the orientation of β-sheets and the presence of an additional β-sheet region in K12111 that replaces a turn region found in K01190. These variations in

β-sheet regions contribute to the divergence observed, resulting in relatively low sequence identity (37%) in the PDB comparison using the FATCAT flexible algorithm (Ye & Godzik, 2004). Nevertheless, a TM-score of 0.87 supports the structural homology between these representatives, suggesting significant similarities despite the noted differences (Hung & Samudrala, 2012; Xu & Zhang, 2010).

The glycosyl hydrolase family 2 (CAZy GH2)., CAZy database catalog(Lombard et al., 2014), encompasses a diverse array of enzymes including arabinopyranosidases, β-1,3-galactosidases, β-galacturonidases, lactase, glycyrrhizin β-glucuronidase, β-D-galactofuranosidase, mannosylglycoprotein endo-β-mannosidases, exo-β-1,4-glucosaminidases, β-glucosidases, β-galactosidases, among others (Drula et al., 2022) This diversity supports the proposition of a medullar structure and suggests a common homology across a shared structural domain. This domain appears to have diversified through at least four distinct evolutionary paths, indicating a likely polyphyletic origin of the catalytic domain.

Moreover, this comparison enable to recognize specific regions, namely Glyco_hydro_2_N and Glyco_hydro_2_C, which exhibit a lower level of homology limitated to certain regions. If we use the sequence of a particular representative of K01190 and is compared with K12111 although the percent of identity is limited into certain positions and amino acids regions (Query coverate of 90% and only a percent of identity that oscilated among, 30 and 40% of identity, if we compare them with BLAST webpage version). Despite this lower homology, these regions maintain a similar orientation and distribution of β-sheets, suggesting a conserved structural role ~~como se demuestra?.~~ The presence of β-strand regions identified by Pfam as lacZ_4 and Bgal_small_N in these structures introduces variable features across the analyzed tertiary structures. This first comparison draws attention to distinctive elements that may be common across both conserved and derived regions.

Utilizing the PDB pairwise comparison tool (Zardecki et al., 2022), is possible to assert these elements are structurally homologous and delineate a specific pattern that defines this group as a

distinct structural gene family when PDB pairwise comparison tool was used (Zardecki et al., 2022),. This finding shows some insights related to the evolutionary and functional dynamics within this enzyme family.

In the effort to validate the identification using sequence analysis via the PDB website and additionally to link both groups, structure and Pfam motifs, we identified two crystal structures that persist despite variations in the sequences used by groups K12308 and K12309. This discovery initially suggests the likelihood of two distinct structures might represent~~ing~~ conserved motifs and structures across both KO groups.~~.~~

The first structure, 1TG7 from a *Penicillium* sp. (Rojas et al., 2004) was compared to 4MAD, a structure derived from a heterologous protein from *Niallia circulans* (formerly *Bacillus*) (Henze et al., 2014) . Initially, we anticipated potential homology between these models based on PDB search results and the visual similarity; however, the alignment of the GH35 catalytic domain with ABD1 and ABD2 proved inconclusive. This comparison yielded a TM-score of 0.44 and a low identity percentage of 23%, indicating a negative result for structural alignment.

Upon further examination of sequences and structural motifs within both KO groups, it was possible to discern a pattern matching the motifs found in the 4MAD structure. This pattern corresponds to the shared signal identified through Pfam motifs in both K12308 and K12309 (Figure 4). This analysis suggests that these two groups may share homology at the primary structure level and correspond to regions found in 4MAD. Consequently, the alternative structure, 1TG7, was excluded from further analysis based on the PDB sequence search. Is in this process where is important to note that both,the comparison of Pfam motifs and tertiary structure comparison and alignment could ~~brin and~~ adequate determination of structural similarity recognizing functional components.

The 4MAD was selected as a model for this gene family, which seems to be consistent across Proteobacteria and represents a common structure in sequences obtained from eukaryotic genomes. This model supports the idea that these KO groups could potentially be unified into a single structural group, emphasizing the relevance of 4MAD in understanding the conserved features within this family. This negative result enable to recognize that beside the probable recognition of limited regions of convergence and similarity, is possible to detect that both the incidence of conserved Pfam motifs and the structural comparison.

Using the Maximum Likelihood phylogenetic tree generated by the web version of IQ-TREE (Nguyen et al., 2015), we could assess the conservation and arrangement of structures within a diverse group of sequences. These sequences included at least three different types of enzymes, each exhibiting unique variations in folding and primary sequence that the Pfam search algorithm did not initially recognize.

As shown in Figure 5, and building on the insights from Figure 1, K01224 emerges as the most diverse among the KO groups. Our analysis of this group via Pfam initially yielded ambiguous results due to the overrepresentation of various motifs. Therefore, we opted to rely on CAZy motifs, which were detected automatically and helped clarify the specificity and recognition of subunits and regions within the diverse sequences analyzed.

Specifically, in the case of group K01224, we observed a complex arrangement of glycosyl hydrolases, each featuring distinct structural configurations. This diversity extends to the presence of elements that exhibit and facilitate the recombination of GH53 and CBM61 motifs, both identified through the CAZy database. This discovery underscores the practical implications of using CAZy for identifying homology among different enzyme representatives, enhancing our understanding of enzyme function and structural diversity.

This approach allowed to propose homology among different representatives based on functional and structural analysis, illustrating the complex interplay and evolutionary relationships within this enzyme family.

The presence of GH53 in the analyzed groups provides crucial insights into the incidence and homology associated with the structural domain, particularly the α/β 8-barrel in the catalytic domain, which shows a diverse evolutionary origin. The combination of GH53 and CBM61 motifs contributes significantly to the diversity and observed duplications among different representatives. Specifically, the distribution of CBM61 highlights the variety of structural arrangements and potential duplications within these proteins.

As depicted in Figure 5 , we present a diverse group of arabinogalactan-β-galactosidases, characterized by a particularly diverse motif structure among representatives. This diversity is evident even when comparing only the distinct elements from the Bacillota and Actinomycetota with those from Thermotoga representatives, suggesting a broad range of structural variations across different taxonomic groups.

Figure 5 presents an arrangement of the diversity within KO group K01224 with the diversification and conserved regions resulting from duplication and subfunctionalization processes. Pfam reports a complete GH53 shared domain for this group, allowing us initially to infer a common structural and functional signal related to previous KO groups, specifically Glyco_hydro_53. However, a more detailed analysis using the HMMER tool provided additional data, revealing at least nine different models characterized by specific structural and functional motifs.

These nine distinct arrangements allow for a comparative analysis that introduces varied representation and arrangements into the phylogenetic analysis results, as presented in Figure 5.

The duplication of CBM61 motifs at both the N and C terminal regions. This feature adds significant insight into the arrangement presented in the phylogenetic tree.

While a common motif, GH53, correlated with the catalytic domain, is recognized, there is also a prevalent incidence of the CBM61 motif. This motif is predominantly reported in taxa such as Bacillota and Actinomycetes, as well as in groups like FCB, Thermotogales, and Dictyoglomus, among other phylogenetically related groups. A study by Cid et al. (2010) identified this motif's role in Beta-1,4 galactanase activity. Additionally, structural analysis underscores the importance of this region in substrate binding, specifically through a mode that resembles starch recognition.

The presence of this motif across all the family indicates a ubiquitous distribution, but the way it interacts with galactan and starch introduces a significant element of diversity. This interaction not only defines the functional landscape of these enzymes but also impacts their evolutionary trajectories by facilitating diversification through structural variation and motif recombination. This comprehensive understanding of motif distribution and function helps elucidate the complex evolution and specialization of enzymes within this family, aligning structural data with phylogenetic insights to map the functional dynamics of these diverse enzymes.

In Figure 5, it is evident that thermophilic and hyperthermophilic sequences, highlighted by the tips and branches colored in red and orange, exhibit unique variations. This distinctive coloring helps to visually categorize these sequences according to their thermal adaptation characteristics. Further, during our sequence comparison, we observed a notable incidence of subfunctional copies of beta-galactosidase within the genus *Alicyclobacillus*. This finding has not been previously reported, indicating a new phenomenon observed in some representatives of these species.

Additionally, the distribution of the CBM61 motif presents a significant aspect for further study. Notably, we observed the presence of this motif at the N-terminal in the genus *Cellulosilyticum*, and

interestingly, it appears duplicated in species not associated with an extremophilic lifestyle. This observation prompts a deeper examination of these domains to propose a probable pathway of past diversification.

Figure 6, presents the results from sequence comparisons aligned with a focus on conserving motif signals, we were able to relate these sequence blocks effectively. This alignment process emphasizes the importance of maintaining a stringent criterion for identifying and filtering conserved motifs across the sequences. To use such targeted alignments, we enhance our ability to discern evolutionary trends and functional implications of these motifs within the enzyme family.

This approach not only supports the identification of structural and functional diversity within beta-galactosidases but also aids in understanding the evolutionary adaptations that contribute to the enzyme's functionality across different environmental conditions and taxonomic groups. This comprehensive analysis highlights the intricate relationship between sequence variation, structural motifs, and enzymatic function, providing deeper insights into the evolutionary dynamics of these enzymes.

Figure 6 builds upon our ongoing study to analyze the CBM61 motif, specifically observing duplications and shared patterns in sequences from hyperthermophilic microorganisms. Through the use of UCSF Chimera, we were able to highlight shared structural elements, coloring them in pink for emphasis, and indicating semi-conservative regions using a color scale from pale green (lower conservation) to magenta (high conservation). Additionally, cluster analysis conducted using R with tools like *cluster*, *hclust*, and *factoextra*, enabled us to infer a shared distribution of amino acids across these sequences.

The multiple sequence alignment illustrated in Figure 6 revealed conserved positions that are crucial for interactions with branched carbohydrates and subunits. This alignment allowed us to identify specific amino acids—phenylalanine, tyrosine, and tryptophan—conserved across different

sequences. These residues play a crucial role in recognizing the lower and central regions of the beta-strands, which are essential for surface interactions, folding, and binding to substrates.

Further examination revealed that the N-terminal CBM61 exhibits a distinct pattern and signal comparable to those from hyperthermophilic origins, while the duplicated motifs display similar proportions suggesting at least three divergent clusters in these sequences. This analysis hints at a unique signal in *Thermotoga* species, which converges only with another similar amino acid content sequence from the *Bacillota*, despite both being hyperthermophilic. This observation suggests at least three potential evolutionary trends within these motifs, underpinning a path of convergent and divergent evolutions.

In light of findings within KO group K01224 regarding structure, conserved positions, and specific amino acid changes, we observed that KO group K01220, predominantly found in Gammaproteobacteria and particularly in Enterobacteria, shares similarities. This group primarily exhibits the GH1 motif identified by CAZy, suggesting a catalytic domain akin to those seen in K01190 and K12111 but differs by complementing specific features in an additional evolutionary branch. This supports the initial proposal that GH1 and GH2 could represent variations of the α/β 8-barrel catalytic motif, demonstrating how these enzyme groups share foundational structural elements while diverging in certain functional capacities. This comparative analysis illuminates the structural and functional diversity within these enzyme families and underscores the importance of conserved motifs in understanding their evolutionary biology and substrate interactions.

Discussion

The methodology employed in linking diverse enzyme groups by a common reaction—specifically, beta-galactosidase activity—highlights a straightforward approach to exploring evolutionary relationships within the KEGG Orthology (KO) framework. This method facilitates the

identification of probable divergence events and the emergence of independent lineages within the same metabolic pathway, which may have originated at different times and within distinct phylogenetic groups.

By simplifying the approach, we can discern potential modifications, such as those between KO12111 and K01190. This insight reveals unique beta-sheet domains that may play critical roles in subunit interactions, and in recognizing alternative regions involved in a single fibrillar molecule specialized for specific bond-breaking interactions and other degradation-related activities (Figure 2). Furthermore, pairwise comparisons enhance our understanding of these structures, revealing instances where there may be no reported sequence homology, yet the TM values and identity percentages suggest new avenues for investigation, albeit with caution.

However, the use of sequence search algorithms like those on the PDB page (Zardecki et al., 2022) sometimes fails to detect certain elements of our specifically analyzed samples obtained from KEGG. Despite these limitations, we can observe both positive and negative results in KO group fusion and structural integration, as proposed by the tools available through BLAST, structural alignment, and the shared presence of Pfam and CAZy motifs.

~~Our~~The focus on linking disparate features into a common structural framework is aimed at analyzing identified signals to recognize a shared motif that may support elements involved in the conservation across three cellular domains (Figure 4). Although preliminary, these comparisons may serve as an initial approach to integrating KO groups and orthologous databases across different genes. This effort seeks to recognize shared distributions, the limits and reach of which may support broader evolutionary theories, including the local ancestry of major taxonomic groups in bacteria.

This analysis potentially aligns with the phylogenetic framework proposed by Gupta (Gupta, 2000), enhancing arguments for a shared Darwinian view of prokaryotic diversity. Such perspectives

underscore the visibility of all branches within the universal tree through current bioinformatic data, suggesting a rich tapestry of evolutionary history encoded within metabolic pathways and structural domains across diverse bacterial lineages.

In ~~our~~this study, for instance, ~~we~~is observed subfunctionalization and gene fusion events that complicate the evolutionary narrative. Such complexities are exemplified in genera like *Alicyclobacillus*, where at least two copies of a gene may possess a distinct catalytic motif, as depicted in Figure 5. This finding raises pertinent questions about the presence of specific motifs associated with thermophilic features and the duplication of genes for specific enzymes.

The duplication and recombination of motifs within the KO group K01224, for instance, illustrates how different domains can coalesce to form a new functional enzyme, such as arabinogalactan beta-galactosidase. The commonality of beta-sheet motifs alongside the α/β 8-barrel domain in families GH2, GH53, or GH35, points to an evolutionary strategy where beta-sheets stabilize or complement the enzymatic function of beta-galactosidases. Motifs like CBM32 and CBM13, which are linked to carbohydrate-binding and are found in various glycoside hydrolases and transferases, underscore the diversity within these enzyme families and hint at a rich tapestry of functional diversification and structural stability (CAZy database).

The motif diversity and the dynamics of gene duplication in prokaryotic genomes reveal a continually evolving landscape where metabolic functions are refined or altered through gene rearrangements, recombination, and potentially horizontal gene transfer. Such processes challenge our understanding of what constitutes "complete" genetic units, as comparative genomic studies often reveal more intact copies of non-housekeeping genes, suggesting a complex scenario of gene retention and functional divergence (Zhang et al., 2015).

Future research should focus on integrating multiple data sources and employing various analytical strategies to construct a more comprehensive view of gene family evolution. For instance,

recognizing conserved amino acids and assessing the structural roles of beta-sheets could lead to insights into enzyme function and stability. Modifications at specific sites, especially in thermophilic enzymes, might enhance substrate recognition and enzymatic efficiency, contributing to our understanding of enzyme adaptation to extreme environments.

By continuously refining our analytical methods and embracing a more holistic view of genomic data, we can better understand the subtleties of enzyme evolution, the role of structural motifs in functional adaptation, and the broader implications of gene duplication and diversity. This comprehensive approach will enable us to not only identify new homologous groups but also develop more stable and efficient enzymes through in silico modeling and molecular docking studies, ultimately pushing the boundaries of our knowledge in enzyme functionality and evolution.

The shared composition and analysis proposed in all the comparison that represent beta-galactosidases enables to recognize the diversity of hydrolases and still this comparison enable to detect possible protein modules with biotechnological importance which bring possible modification in Bacillota homologues which bring probable modification into models focused into similar elements in CBM61 thermophilic structural models.

Conclussion:

The recognition of similar and representative structural elements involved in protein structure fucntionality bring the idea of convergence or homology in the path to recognize and link the previously orthologous groups.The implementation to combine KEGG databases as a primer and combine the CAZy conserved specialized motifs in carbohydratases bring the sufficient data and material to recognize eventsof subfunctional duplication.The implementation of this method bring a dual way in recognize if this elements are the result of differential interactions whose can define structural elements who share a evolutionary story and based on structural components can be defined possible functional elements involved in thermostability. Combining tertiary and previously

motifs reported by different databases bring a dual (Pfam and CAZy) control that bring robustness in the comparison of different orhologous groups. Additionally the identification of certain conserved motifs like CBM61 enable to recognize shared and particular regions that can be used in later biothechnological structure and modifications.

References

Altschul, S. (1997). Gapped BLAST and PSI-BLAST: A new generation of protein database search programs. *Nucleic Acids Research*, *25*(17), 3389-3402. https://doi.org/10.1093/nar/25.17.3389

Anisha, G. S. (2016). β-Galactosidases. En *Current Developments in Biotechnology and Bioengineering: Production, Isolation and Purification of Industrial Products*. Elsevier Inc. https://doi.org/10.1016/B978-0-444-63662-1.00017-8

Audic, S., & Claverie, J. M. (1998). Self-identification of protein-coding regions in microbial genomes. *Proceedings of the National Academy of Sciences of the United States of America*, *95*(17), 10026-10031.

Becerra, A., Delaye, L., Lazcano, A., & Orgel, L. E. (2007). Protein disulfide oxidoreductases and the evolution of thermophily: Was the last common ancestor a heat-loving microbe? *Journal of Molecular Evolution*, *65*(3), 296-303. https://doi.org/10.1007/s00239-007-9005-0

Brüschweiler, R. (2002). Efficient RMSD measures for the comparison of two molecular ensembles. *Proteins: Structure, Function, and Bioinformatics*, *50*(1), 26-34. https://doi.org/10.1002/prot.10250

Chan, Y. H., Venev, S. V., Zeldovich, K. B., & Matthews, C. R. (2017). Correlation of fitness landscapes from three orthologous TIM barrels originates from sequence and structure constraints. En *NATURE COMMUNICATIONS* (Vol. 8). NATURE PUBLISHING GROUP. https://doi.org/10.1038/ncomms14614

Cid, M., Pedersen, H. L., Kaneko, S., Coutinho, P. M., Henrissat, B., Willats, W. G. T., & Boraston, A. B. (2010). Recognition of the helical structure of β-1,4-galactan by a new family of

carbohydrate-binding modules. En *Journal of Biological Chemistry* (Vol. 285, Número 46, pp. 35999-36009). https://doi.org/10.1074/jbc.M110.166330

Collins, R. E., Merz, H., & Higgs, P. G. (2011). Origin and evolution of gene families in Bacteria and Archaea. *BMC Bioinformatics*, *12*(S9), S14. https://doi.org/10.1186/1471-2105-12-S9-S14

Cuneo, M. J., Tian, Y., Allert, M., & Hellinga, H. W. (2008). The backbone structure of the thermophilic Thermoanaerobacter tengcongensis ribose binding protein is essentially identical to its mesophilic E. coli homolog. *BMC Structural Biology*, *8*, 20. https://doi.org/10.1186/1472-6807-8-20

Demuth, J. P., & Hahn, M. W. (2009). The life and death of gene families. *BioEssays : news and reviews in molecular, cellular and developmental biology*, *31*(1), 29-39. https://doi.org/10.1002/bies.080085

Drula, E., Garron, M.-L., Dogan, S., Lombard, V., Henrissat, B., & Terrapon, N. (2022). The carbohydrate-active enzyme database: Functions and literature. *Nucleic Acids Research*, *50*(D1), D571-D577. https://doi.org/10.1093/nar/gkab1045

Dunker, A. K., & Lathrop, R. H. (1995). Protein structure prediction. Introduction to session. *Proceedings of the Hawaii International Conference on System Sciences*, *5*, 295. https://www.scopus.com/inward/record.uri?eid=2-s2.0-0028959050&partnerID=40&md5=849ed154fb6ad6a4103086ec1be4c0ad

Eirín-López, J. M., Rebordinos, L., Rooney, A. P., & Rozas, J. (2012). The Birth-and-Death Evolution of Multigene Families Revisited. En M. A. Garrido-Ramos (Ed.), *Genome Dynamics* (Vol. 7, pp. 170-196). S. Karger AG. https://doi.org/10.1159/000337119

Finn, R. D., Clements, J., Arndt, W., Miller, B. L., Wheeler, T. J., Schreiber, F., Bateman, A., & Eddy, S. R. (2015). HMMER web server: 2015 update. *Nucleic Acids Research*, *43*(W1), W30-W38. https://doi.org/10.1093/nar/gkv397

Finn, R. D., Mistry, J., Tate, J., Coggill, P., Heger, A., Pollington, J. E., Gavin, O. L., Gunasekaran, P., Ceric, G., Forslund, K., Holm, L., Sonnhammer, E. L. L., Eddy, S. R., Bateman, A., El-Gebali, S., Mistry, J., Bateman, A., Eddy, S. R., Luciani, A., … Finn, R. D. (2010a). The

Pfam protein families database in 2019. *Nucleic acids research*, *38*(8), 1-6. https://doi.org/10.1093/nar/gky995

Finn, R. D., Mistry, J., Tate, J., Coggill, P., Heger, A., Pollington, J. E., Gavin, O. L., Gunasekaran, P., Ceric, G., Forslund, K., Holm, L., Sonnhammer, E. L. L., Eddy, S. R., Bateman, A., El-Gebali, S., Mistry, J., Bateman, A., Eddy, S. R., Luciani, A., … Finn, R. D. (2010b). The Pfam protein families database in 2019. *Nucleic acids research*, *38*(8), 1-6. https://doi.org/10.1093/nar/gky995

Forterre, P., Gribaldo, S., & Brochier, C. (2005). [Luca: The last universal common ancestor]. *Médecine sciences : M/S*, *21*(10), 860-865. https://doi.org/10.1051/medsci/20052110860

Glansdorff, N. (2000). About the last common ancestor, the universal life-tree and lateral gene transfer: A reappraisal. *Molecular microbiology*, *38*(2), 177-185.

Gogarten, J. P., & Olendzenski, L. (1999). Orthologs, paralogs and genome comparisons. *Current opinion in genetics & development*, *9*(6), 630-636.

Gupta, R. S. (2000). The Natural Evolutionary Relationships among Prokaryotes. *Critical Reviews in Microbiology*, *26*(2), 111-131. https://doi.org/10.1080/10408410091154219

Hall, B. G. (1999). Experimental evolution of Ebg enzyme provides clues about the evolution of catalysis and to evolutionary potential. *FEMS Microbiology Letters*, *174*(1), 1-8. https://doi.org/10.1111/j.1574-6968.1999.tb13542.x

He, R., Zhang, J., Shao, Y., Gu, S., Song, C., Qian, L., Yin, W.-B., & Li, Z. (2023). Knowledge-guided data mining on the standardized architecture of NRPS: Subtypes, novel motifs, and sequence entanglements. *PLOS Computational Biology*, *19*(5), e1011100. https://doi.org/10.1371/journal.pcbi.1011100

Henikoff, S., Greene, E. A., Pietrokovski, S., Bork, P., Attwood, T. K., & Hood, L. (1997). Gene Families: The Taxonomy of Protein Paralogs and Chimeras. *Science*, *278*(5338), 609-614. https://doi.org/10.1126/science.278.5338.609

Holm, L. (2022). Dali server: Structural unification of protein families. *Nucleic Acids Research*, *50*(W1), W210-W215. https://doi.org/10.1093/nar/gkac387

Hung, L.-H., & Samudrala, R. (2012). Accelerated protein structure comparison using TM-score-GPU. *Bioinformatics*, *28*(16), 2191-2192. https://doi.org/10.1093/bioinformatics/bts345

Innan, H., & Kondrashov, F. (2010). The evolution of gene duplications: Classifying and distinguishing between models. *Nature reviews. Genetics*, *11*(2), 97-108. https://doi.org/10.1038/nrg2689

Juers, D. H., Matthews, B. W., & Huber, R. E. (2012). LacZ β-galactosidase: Structure and function of an enzyme of historical and molecular biological importance. *Protein Science : A Publication of the Protein Society*, *21*(12), 1792-1807. https://doi.org/10.1002/pro.2165

Kanehisa. (1996). *Kyoto Encyclopedia of Genes and Genomes*.

Kanehisa, M. (2000). KEGG: Kyoto Encyclopedia of Genes and Genomes. *Nucleic Acids Research*, *28*(1), 27-30. https://doi.org/10.1093/nar/28.1.27

Kanehisa, M., Goto, S., Kawashima, S., Okuno, Y., & Hattori, M. (2004). The KEGG resource for deciphering the genome. *Nucleic acids research*, *32*(Database issue), D277-80. https://doi.org/10.1093/nar/gkh063

Kassambara, A. (2017). *Practical Guide To Cluster Analysis in R Unsupervised Machine Learning*. CreateSpace Independent Publishing Platform. https://doi.org/978-1542462709

Kondrashov, F. A., Rogozin, I. B., Wolf, Y. I., & Koonin, E. V. (2002). Selection in the evolution of gene duplications. *Genome biology*, *3*(2), RESEARCH0008.

Konrad, A., Teufel, A. I., Grahnen, J. A., & Liberles, D. A. (2011). Toward a general model for the evolutionary dynamics of gene duplicates. *Genome biology and evolution*, *3*, 1197-1209. https://doi.org/10.1093/gbe/evr093

Koonin, E. V. (2005). Orthologs, paralogs, and evolutionary genomics. *Annual review of genetics*, *39*, 309-338. https://doi.org/10.1146/annurev.genet.39.073003.114725

Kristensen, D. M., Wolf, Y. I., Mushegian, A. R., & Koonin, E. V. (2011). Computational methods for Gene Orthology inference. *Briefings in bioinformatics*, *12*(5). https://doi.org/10.1093/bib/bbr030

Li, S. C. (2013). The difficulty of protein structure alignment under the RMSD. *Algorithms for Molecular Biology*, *8*(1), 1. https://doi.org/10.1186/1748-7188-8-1

Liu, Y., Makarova, K. S., Huang, W.-C., Wolf, Y. I., Nikolskaya, A. N., Zhang, X., Cai, M., Zhang, C.-J., Xu, W., Luo, Z., Cheng, L., Koonin, E. V., & Li, M. (2021). Expanded diversity of

Asgard archaea and their relationships with eukaryotes. *Nature*, *593*(7860), 553-557. https://doi.org/10.1038/s41586-021-03494-3

Lombard, V., Golaconda Ramulu, H., Drula, E., Coutinho, P. M., & Henrissat, B. (2014). The carbohydrate-active enzymes database (CAZy) in 2013. *Nucleic Acids Research*, *42*(D1), D490-D495. https://doi.org/10.1093/nar/gkt1178

Lynch, M., & Conery, J. S. (2000). The Evolutionary Fate and Consequences of Duplicate Genes. *Science*, *290*(5494), 1151-1155. https://doi.org/10.1126/science.290.5494.1151

Magner, A., Szpankowski, W., & Kihara, D. (2015). On the Origin of Protein Superfamilies and Superfolds. *Scientific Reports*, *5*(1), 8166. https://doi.org/10.1038/srep08166

Mewis, K., Lenfant, N., Lombard, V., & Henrissat, B. (2016). Dividing the Large Glycoside Hydrolase Family 43 into Subfamilies: A Motivation for Detailed Enzyme Characterization. *Applied and Environmental Microbiology*, *82*(6), 1686-1692. https://doi.org/10.1128/AEM.03453-15

Mistry, J., Chuguransky, S., Williams, L., Qureshi, M., Salazar, G. A., Sonnhammer, E. L. L., Tosatto, S. C. E., Paladin, L., Raj, S., Richardson, L. J., Finn, R. D., & Bateman, A. (2021). Pfam: The protein families database in 2021. *Nucleic Acids Research*, *49*(D1), D412-D419. https://doi.org/10.1093/nar/gkaa913

Nei, M., & Rooney, A. P. (2005). Concerted and Birth-and-Death Evolution of Multigene Families. *Annual Review of Genetics*, *39*(1), 121-152. https://doi.org/10.1146/annurev.genet.39.073003.112240

Nguyen, L.-T., Schmidt, H. A., von Haeseler, A., & Minh, B. Q. (2015). IQ-TREE: A Fast and Effective Stochastic Algorithm for Estimating Maximum-Likelihood Phylogenies. *Molecular Biology and Evolution*, *32*(1), 268-274. https://doi.org/10.1093/molbev/msu300

Pandit, S. B., Gosar, D., Abhiman, S., Sujatha, S., Dixit, S. S., Mhatre, N. S., Sowdhamini, R., & Srinivasan, N. (2002). SUPFAM–a database of potential protein superfamily relationships derived by comparing sequence-based and structure-based families: Implications for structural genomics and function annotation in genomes. *Nucleic acids research*, *30*(1), 289-293.

Pettersen, E. F., Goddard, T. D., Huang, C. C., Couch, G. S., Greenblatt, D. M., Meng, E. C., & Ferrin, T. E. (2004). UCSF Chimera—A visualization system for exploratory research and analysis. *Journal of Computational Chemistry*, *25*(13), 1605-1612. https://doi.org/10.1002/jcc.20084

Rigden, D. J., & Fernández, X. M. (2021). The 2021 *Nucleic Acids Research* database issue and the online molecular biology database collection. *Nucleic Acids Research*, *49*(D1), D1-D9. https://doi.org/10.1093/nar/gkaa1216

Rojas, A. L., Nagem, R. A. P., Neustroev, K. N., Arand, M., Adamska, M., Eneyskaya, E. V., Kulminskaya, A. A., Garratt, R. C., Golubev, A. M., & Polikarpov, I. (2004). Crystal Structures of β-Galactosidase from Penicillium sp. And its Complex with Galactose. *Journal of Molecular Biology*, *343*(5), 1281-1292. https://doi.org/10.1016/j.jmb.2004.09.012

Russell, R. B., & Barton, G. J. (1992). Multiple protein sequence alignment from tertiary structure comparison: Assignment of global and residue confidence levels. *Proteins*, *14*(2), 309-323. https://doi.org/10.1002/prot.340140216

Schönknecht, G., Chen, W.-H., Ternes, C. M., Barbier, G. G., Shrestha, R. P., Stanke, M., Bräutigam, A., Baker, B. J., Banfield, J. F., Garavito, R. M., Carr, K., Wilkerson, C., Rensing, S. A., Gagneul, D., Dickenson, N. E., Oesterhelt, C., Lercher, M. J., & Weber, A. P. M. (2013). Gene transfer from bacteria and archaea facilitated evolution of an extremophilic eukaryote. *Science*, *339*(6124), 1207-1210. https://doi.org/10.1126/science.1231707

Shafee, T. M. A., Robinson, A. J., Van Der Weerden, N., & Anderson, M. A. (2016). Structural homology guided alignment of cysteine rich proteins. *SpringerPlus*, *5*(1), 27. https://doi.org/10.1186/s40064-015-1609-z

Simossis, V. A., & Heringa, J. (2005). PRALINE: A multiple sequence alignment toolbox that integrates homology-extended and secondary structure information. *Nucleic Acids Research*, *33*(Web Server), W289-W294. https://doi.org/10.1093/nar/gki390

Thoma, R., Schwander, M., Liebl, W., Kirschner, K., & Sterner, R. (1998). A histidine gene cluster of the hyperthermophile Thermotoga maritima: Sequence analysis and evolutionary significance. *Extremophiles : life under extreme conditions*, *2*(4), 379-389.

Uchil, P. D., Nagarajan, A., & Kumar, P. (2017). β-Galactosidase. *Cold Spring Harbor Protocols*, *2017*(10), pdb.top096198. https://doi.org/10.1101/pdb.top096198

Wang, J., Zhang, C., & Li, Y. (2022). Genome-Wide Identification and Expression Profiles of 13 Key Structural Gene Families Involved in the Biosynthesis of Rice Flavonoid Scaffolds. *Genes*, *13*(3), 410. https://doi.org/10.3390/genes13030410

Wang, S., Guo, G., Li, L., Cao, L., Tong, L., Ren, G., & Liu, Y. (2014). Identification and characterization of an unusual glycosyltransferase-like enzyme with β-galactosidase activity from a soil metagenomic library. En *Enzyme and Microbial Technology* (Vol. 57, pp. 26-35). https://doi.org/10.1016/j.enzmictec.2014.01.007

Wong, W.-C., Maurer-Stroh, S., & Eisenhaber, F. (2011). THE JANUS-FACED E-VALUES OF HMMER2: EXTREME VALUE DISTRIBUTION OR LOGISTIC FUNCTION? *Journal of Bioinformatics and Computational Biology*, *09*(01), 179-206. https://doi.org/10.1142/S0219720011005264

Xu, J., & Zhang, Y. (2010). How significant is a protein structure similarity with TM-score = 0.5? *Bioinformatics*, *26*(7), 889-895. https://doi.org/10.1093/bioinformatics/btq066

Ye, Y., & Godzik, A. (2004). FATCAT: A web server for flexible structure comparison and structure similarity searching. *Nucleic Acids Research*, *32*(Web Server), W582-W585. https://doi.org/10.1093/nar/gkh430

Zardecki, C., Dutta, S., Goodsell, D. S., Lowe, R., Voigt, M., & Burley, S. K. (2022). PDB-101: Educational resources supporting molecular explorations through biology and medicine. *Protein Science*, *31*(1), 129-140. https://doi.org/10.1002/pro.4200

Zaremba-Niedzwiedzka, K., Caceres, E. F., Saw, J. H., Bäckström, D., Juzokaite, L., Vancaester, E., Seitz, K. W., Anantharaman, K., Starnawski, P., Kjeldsen, K. U., Stott, M. B., Nunoura, T., Banfield, J. F., Schramm, A., Baker, B. J., Spang, A., & Ettema, T. J. G. (2017). Asgard archaea illuminate the origin of eukaryotic cellular complexity. *Nature*, *541*(7637), 353-358. https://doi.org/10.1038/nature21031

Zhang, H., Yohe, T., Huang, L., Entwistle, S., Wu, P., Yang, Z., Busk, P. K., Xu, Y., & Yin, Y. (2018). dbCAN2: A meta server for automated carbohydrate-active enzyme annotation. *Nucleic Acids Research*, *46*(W1), W95-W101. https://doi.org/10.1093/nar/gky418

Zhang, X., Peng, C., Zhang, G., & Gao, F. (2015). Comparative analysis of essential genes in prokaryotic genomic islands. *Scientific Reports*, *5*(1), 12561. https://doi.org/10.1038/srep12561

KEGG KO group	Definition	EC number	Catalitic reaction	Genes related

K01190	Beta-galactosidase	3.2.1.23	Hydrolysis of terminal non-reducing	*lacZ, gly2E, gal2C, rhgZ, yesZ,*
K01220	6 Phospho-beta-galactosidase	3.2.1.85	6 phospho beta-galactosdase + H2O → 6 phospho-D galactose + alcohol	*lacG*
K01224	Arabino-galactan endo-1-4-beta galactosidase	3.2.1.89	Hydrolyses th beta (1-4) bonds found in homologues from plants. Produces galactan endo beta 1,3-galactanase	-
K12111	Evolved beta-galactosidase subunit alpha	3.2.1.23	Hydrolysis of terminal non reducing beta – D – galactose residues in beta – D galactosides	*ebgA*
K12112	Evolved beta-galactosidase subunit beta	-	Non-fuctional subunit, involved in the interaction of ebgA, ebgB and ebgC. Not catalytic subunit	*ebgC*
K12308	Beta - galactosidase	3.2.1.23	Hydrolysis of terminal non-reducing beta – D – galactose residues in beta – D galactosides	*bgaB, lacA, bglY*
K12309	Beta - galactosidase	3.2.1.23	Hydrolysis of terminal non reducing beta – D – galactose residues in beta – D - galactosides	*GLB1*
K18579	Galactan endo-1,6-beta-galactosidase	3.2.1.164	Endohydrolysis of 1 – 6 – beta – D - galactosidic linkages in arabino galactan proteins and (1→3): (1→6) beta galactansto yield galactose and (1→6) beta-galactobiose as final products	*6GAL*
K25543	Beta-galactosidase 1-like protein	3.2.1.-	Unknown	*GLB1L*

Table 1. The KO analyzed and related to the beta-galactosidase function. The corresponding description encodes the genes for each one of the cases.

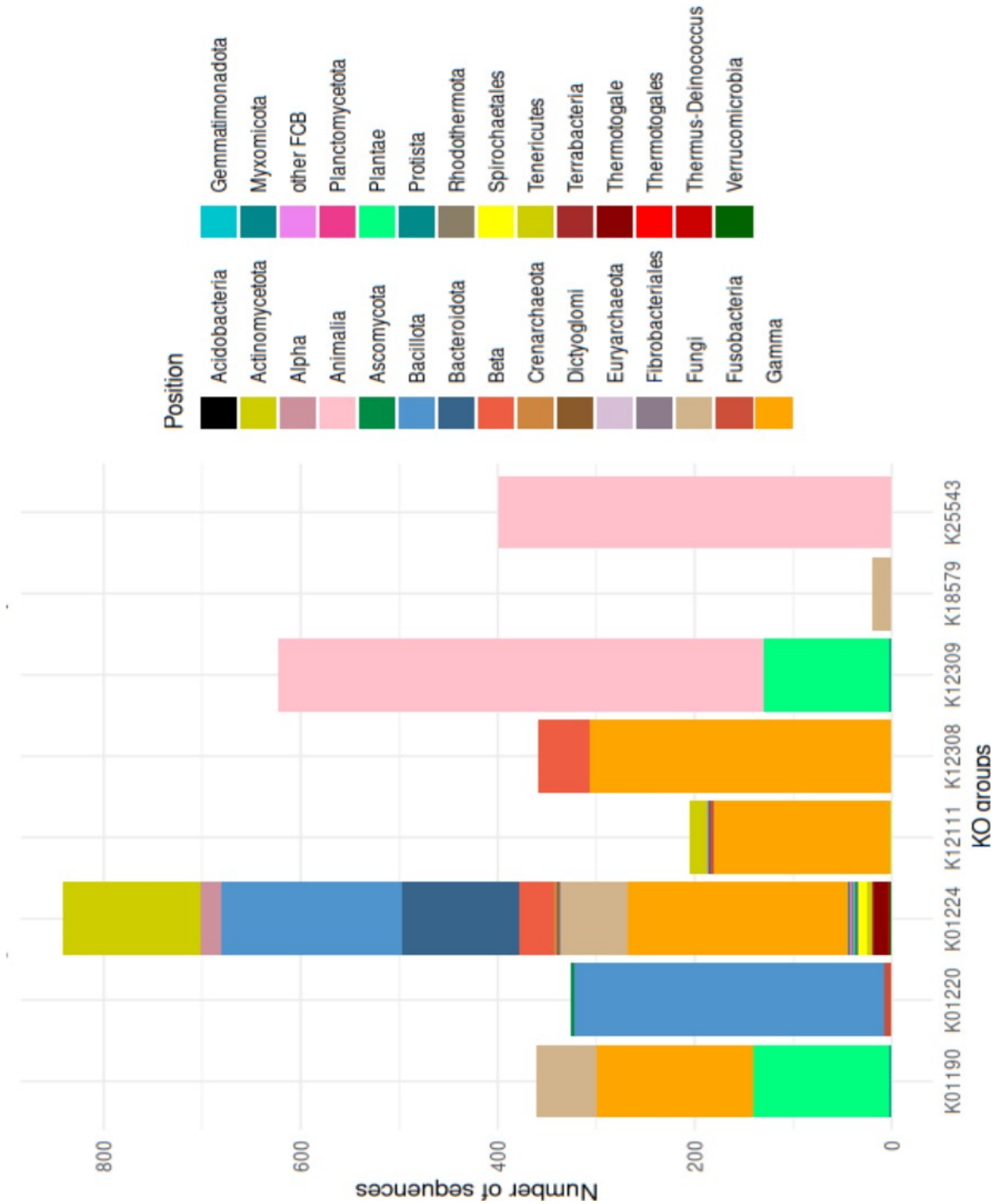
Position
Acidobacteria
Actinomycetota
Alpha
Animalia
Ascomycota
Bacillota
Bacteroidota
Beta
Crenarchaeota
Dictyoglomi
Euryarchaeota
Fibrobacteriales
Fungi
Fusobacteria
Gamma
Gemmatimonadota
Myxomicota
other FCB
Planctomycetota
Plantae
Protista
Rhodothermota
Spirochaetales
Tenericutes
Terrabacteria
Thermotogale
Thermotogales
Thermus-Deinococcus
Verrucomicrobia
Number of sequences
0
200
400
600
800
K01190
K01220
K01224
K12111
K12308
K12309
K18579
K25543
KO groups

Figure 1: Taxonomic distribution of beta-galactosidase reactions across KO groups. This figure illustrates the comparative taxonomic distribution within eight KO groups associated with beta-galactosidase reactions. Each group is labeled to indicate specific subdivisions within the Proteobacteria: Alpha (Alphaproteobacteria), Beta (Betaproteobacteria), and Gamma (Gammaproteobacteria). The FCB group encompasses sequences from species outside the Bacteroidota, Fibrobacterota, and Gemmatimonadota, which are also included in this analysis. Additionally, sequences from the Eukaryotic kingdom are marked, representing the major divisions: Plantae, Animalia, Fungi, and Protista. This visualization aids in understanding the phylogenetic breadth and functional diversity of beta-galactosidase enzymes across both prokaryotic and eukaryotic domains.

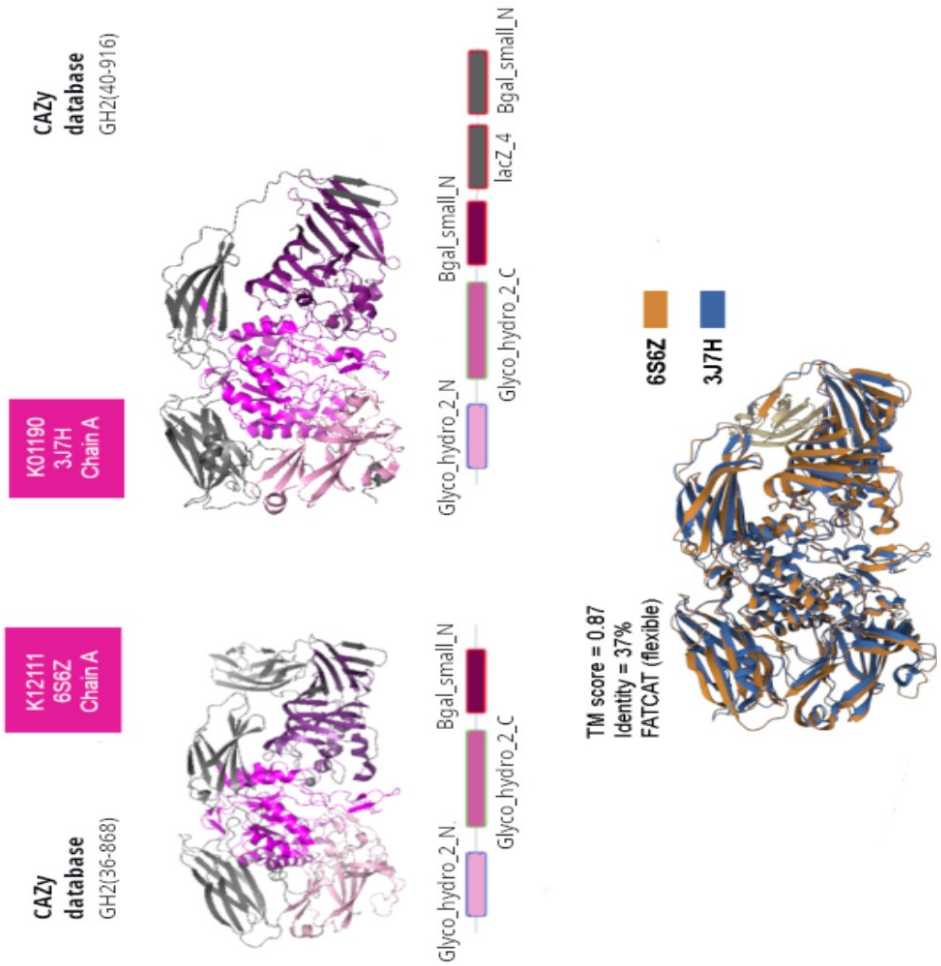

Figure 2: Comparative analysis of orthologous groups K01190 and K12111 of beta glacotisdases 6s6z of *Thermotoga maritima* and 3J7H of *Escherichia coli* . The analysis is conducted using three distinct methods: structural pairwise alignment, Pfam motif comparison, and CAZy database analysis. The lower section of the figure displays the pairwise structural comparison between the two KO groups, highlighting key structural similarities and differences. The results from the structural alignment are detailed on the right side of the figure, providing quantitative data on alignment scores and structural congruence. This comprehensive comparison helps elucidate the evolutionary relationships and functional adaptations within these orthologous groups.

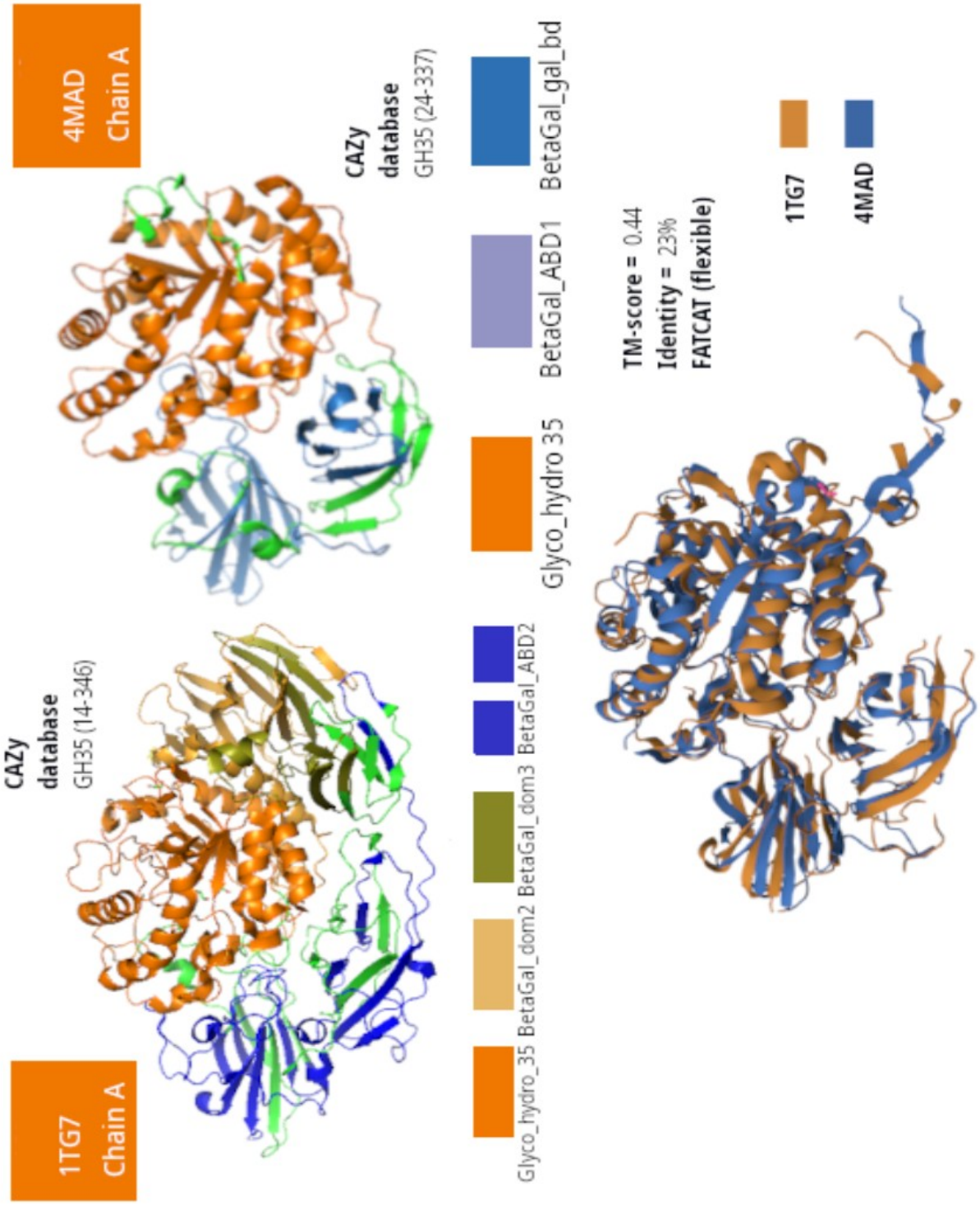
1TG7
Chain A
CAZy
database
GH35 (14-346)
4MAD
Chain A
CAZy
database
GH35 (24-337)
Glyco_hydro_35
BetaGal_dom2
BetaGal_dom3
BetaGal_ABD2
Glyco_hydro 35
BetaGal_ABD1
BetaGal_gal_bd
TM-score = 0.44
Identity = 23%
FATCAT (flexible)
1TG7
4MAD

Figure 3: Tertiary structure comparison of 1TG7 and 4MAD. This figure presents a comparative analysis of the tertiary structures of 1TG7, derived from a *Penicillium* species, and 4MAD, modeled from *Niallia circulans*. Despite their identification as structurally similar by

sequence search tools, this comparison serves as a negative control, demonstrating that common structural incidences do not necessarily imply shared homology. The absence of shared homology is further supported by distinct Pfam recognized motifs, highlighting the importance of motif analysis in confirming structural and functional relationships.

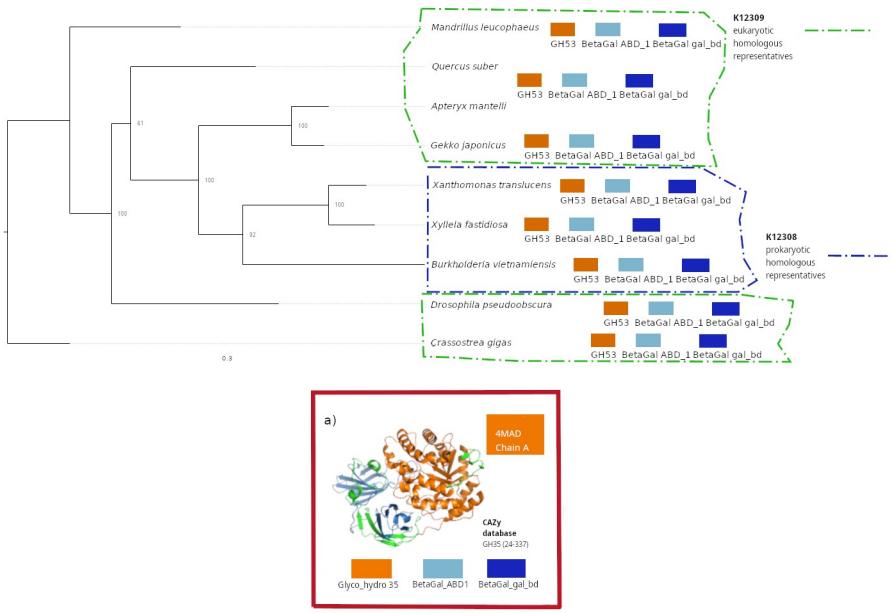
Mandrillus leucophaeus
Quercus suber
Apteryx mantelli
Gekko japonicus
Xanthomonas translucens
Xylella fastidiosa
Burkholderia vietnamiensis
Drosophila pseudoobscura
Crassostrea gigas
GH53 BetaGal ABD_1 BetaGal gal_bd
K12309
eukaryotic
homologous
representatives
K12308
prokaryotic
homologous
representatives
61
100
100
100
100
92
0.3
a)
4MAD
Chain A
CAZy
database
GH35 (24-337)
Glyco_hydro 35
BetaGal_ABD1
BetaGal_gal_bd

Figure 4: Phylogenetic tree of K12308 and K12309 with Pfam motif analysis. This figure displays a phylogenetic tree constructed using representative sequences from the KO groups K12308 and K12309, which are associated with the structure of the beta-galactosidase enzyme modeled after 4MAD (a)). An ultrafast bootstrap method was used to assess the reliability of the phylogenetic tree and highlight the skewed distribution between prokaryotic (K12308) and eukaryotic (K12309) sequences. The tree is rooted with an outgroup sequence from *Crassostrea gigas* (Mollusca) to provide a phylogenetic baseline. Pfam motifs detected in the sequences are annotated throughout the tree to illustrate their prevalence and distribution. Prokaryotic sequences are specifically marked with blue lines in the lower part of the tree, underscoring their phylogenetic placement and relatedness.

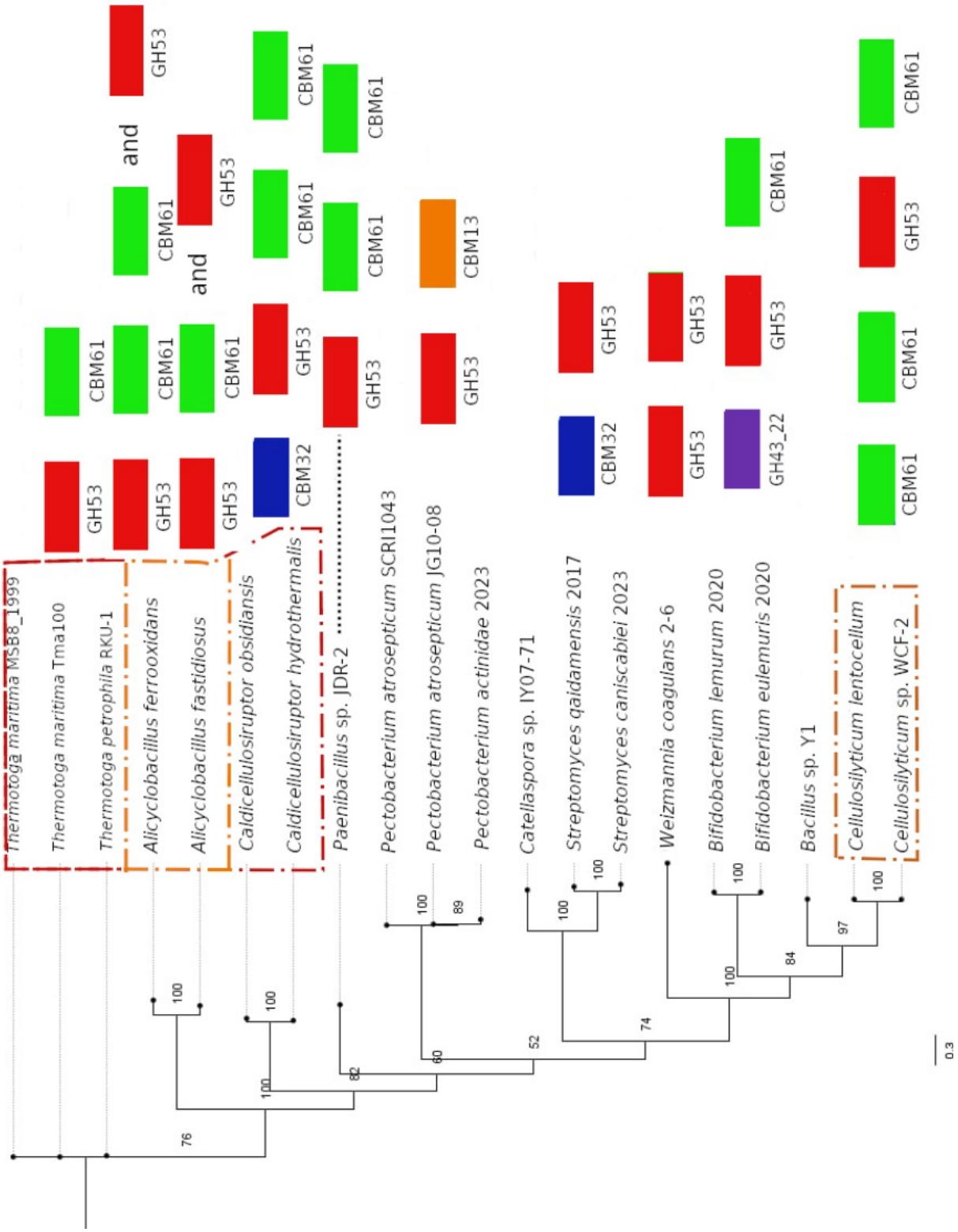
Thermotoga maritima MSB8_1999
Thermotoga maritima Tma100
Thermotoga petrophila RKU-1
Alicyclobacillus ferrooxidans
Alicyclobacillus fastidiosus
Caldicellulosiruptor obsidiansis
Caldicellulosiruptor hydrothermalis
Paenibacillus sp. JDR-2
Pectobacterium atrosepticum SCRI1043
Pectobacterium atrosepticum JG10-08
Pectobacterium actinidae 2023
Catellaspora sp. IY07-71
Streptomyces qaidamensis 2017
Streptomyces caniscabiei 2023
Weizmannia coagulans 2-6
Bifidobacterium lemurum 2020
Bifidobacterium eulemuris 2020
Bacillus sp. Y1
Cellulosilyticum lentocellum
Cellulosilyticum sp. WCF-2
GH53
CBM61
and
CBM32
CBM13
GH43_22
0.3

Figure 5: Phylogenetic tree analysis of K01224 of beta galactosidaseswith CAZy Motif Distribution.-The tree utilizes a Maximum Likelihood approach with ultrafast bootstrap to validate the phylogenetic relationships. Hyperthermophilic and thermophilic species are distinguished with red and orange lines, respectively, to emphasize their thermal adaptation within the tree. The GH53 motif, conserved across all representatives, indicates a shared functional foundation, while the CBM61 domain illustrates occurrences of motif duplication across different branches, pointing to evolutionary diversification. Additional motifs are color-coded and ordered according to their appearance in the primary structure, facilitating an intuitive understanding of motif distribution and its evolutionary implications.

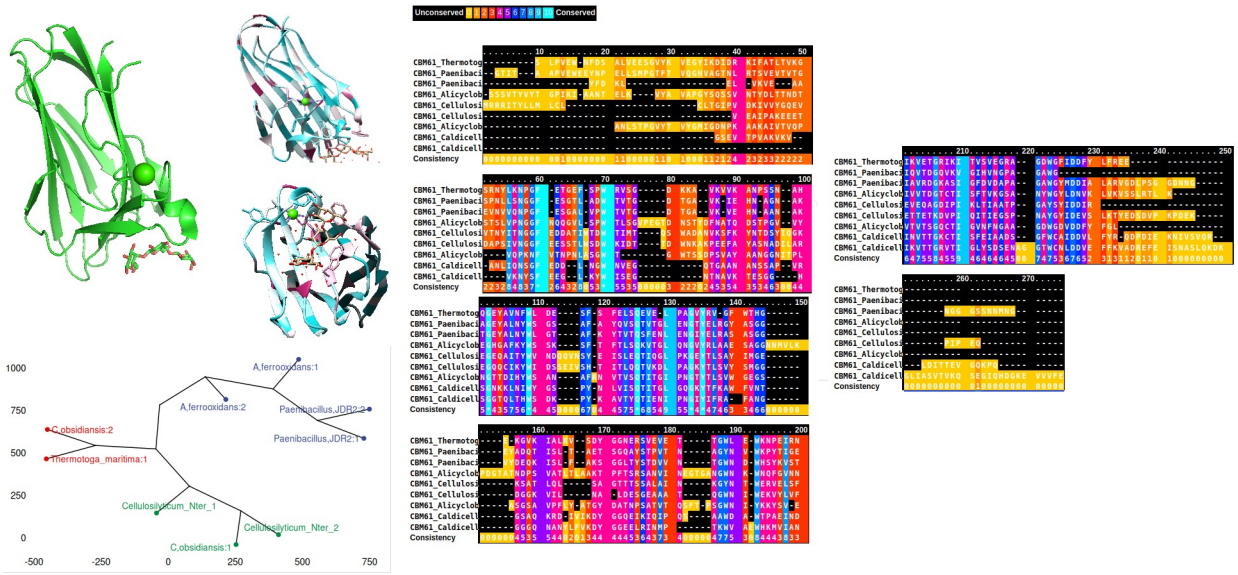
Unconserved
Conserved
CBM61_Thermotog
CBM61_Paenibaci
CBM61_Alicyclob
CBM61_Cellulosi
CBM61_Caldicell
Consistency
A.ferrooxidans:1
A.ferrooxidans:2
Paenibacillus.JDR2:2
Paenibacillus.JDR2:1
C.obsidiansis:2
Thermotoga_maritima:1
Cellulosilyticum_Nter_1
Cellulosilyticum_Nter_2
C.obsidiansis:1
1000
750
500
250
0
-500
-250
0
250
500
750

Figure 6: Comparative Analysis of CBM61 Motif in *Thermotoga maritima* tertiary structure *and homologues*. This figure presents a detailed comparison of the CBM61 motif from *Thermotoga maritima* (shown in green, 2XOM.pdb) using UCSF Chimera. Conserved amino acid positions are highlighted in pink and magenta, emphasizing the most conserved positions across the sequences. Integrated within the figure is a multiple sequence alignment of all CBM61 sequences, with sequence names abbreviated to represent their respective positions in the primary sequence order. Additionally, the amino acid content analysis is visualized through a common scale, leading to a cluster analysis-based dendrogram that categorizes the sequences into three distinct clusters. This clustering relates back to the evolutionary relationships and motif distributions discussed in Figure 5, providing a cohesive view of sequence variation and conservation within the CBM61 motif across different sequences.